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Inventor(s)	Kim; Jin Hyun et al.

Pedal simulator of vehicle

Abstract

The present disclosure relates to a pedal simulator of a vehicle, the pedal simulator includes a piston part slidably disposed in the housing part, an elastic part elastically supporting the piston part within the housing part, a damper part provided in the piston part and compressed onto the housing part by a contact, and a hollow cap part combined with the piston part and interposed between the piston part and the damper part.

Inventors: **Kim; Jin Hyun (Yongin-si, KR), Cha; Myeon Gyu (Yongin-si, KR)**

Applicant: **HYUNDAI MOBIS CO., LTD.** (Seoul, KR)

Family ID: **94875768**

Assignee: **Hyundai Mobis Co., Ltd. (Seoul, KR)**

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References Cited

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2022/0332296	12/2021	Eriksen	N/A	G05G 1/46
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FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
10-2223847	12/2020	KR	N/A

Primary Examiner: Johnson; Vicky A

Attorney, Agent or Firm: Morgan, Lewis & Bockius LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from and the benefit of Korean Patent Application No. 10-2023-0129131, filed on Sep. 26, 2023, which is hereby incorporated by reference for all purposes as if set forth herein.

BACKGROUND

Field

(2) Exemplary embodiments of the present disclosure relate to a pedal simulator of a vehicle, and more particularly, to a pedal simulator of a vehicle that may provide a braking sense.

Discussion of the Background

(3) In general, a hydraulic system has been applied to an electro mechanical brake. Recently, as technologies related to a brake-by-wire system and an autonomous vehicle are emerging, a non-hydraulic brake system needs to be developed.

(4) A pedal simulator is a part that is mounted on an electro mechanical brake or an electronic booster (VEB), and provides a driver with a braking sense that is generated from the existing mechanical (hydraulic) brake.

(5) A conventional pedal simulator uses a plurality of springs and dampers in order to provide a braking sense like a mechanical booster. However, the conventional pedal simulator has problems with an increase of the number of parts and an increase of shape complexity that are required to increase similarity with the mechanical booster.

(6) The Background technology of the present disclosure is disclosed in Korean Patent No. 10-2223847 (issued on Mar. 8, 2021 and entitled “PEDAL SIMULATOR”).

SUMMARY

(7) Various embodiments are directed to providing a pedal simulator of a vehicle, which has an optimized structure and a simplified shape.

(8) Furthermore, various embodiments are directed to providing a pedal simulator of a vehicle, which may be applied regardless of the type and shape of a pedal.
(9) In an embodiment, a pedal simulator of a vehicle may include a housing part, a piston part slidably disposed in the housing part, an elastic part elastically supporting the piston part within the housing part, a damper part provided in the piston part and compressed onto the housing part by a contact, and a hollow cap part combined with the piston part and interposed between the piston part and the damper part.
(10) The piston part may include a piston body part located within the housing part and provided with a ball part, a piston pressurization part mounted on one side of the piston body part and provided with a socket part rotatably combined with the ball part, and a piston rod part mounted on a second side of the piston body part and configured to have the damper part and the cap part mounted thereon.
(11) The piston rod part may be provided with an opening at an end of one side thereof, and may be formed in a hollow form.
(12) The damper part may be received within the piston rod part and may include an elastically deformable material.
(13) The cap part may include a contact part coming in contact with a first part of the damper part to restrict the deformation of the first part, a non-contact part spaced apart from a second part of the damper part to allow the deformation of the second part, and a flange part mounted on an end of one side of the piston rod part.
(14) The housing part may include a hollow part in which the piston part is movably received, a screw part that is screw-combined with the hollow part and includes a middle room part with which the damper part comes into contact or does not come into contact depending on a moving direction of the piston part, and the guide part that is mounted on the outside of the middle room part and communicates with the hollow part, and into which the piston rod part is inserted.
(15) The diameter of the piston body part may be formed to be greater than the diameter of the piston rod part.
(16) The elastic part may have one side that comes into contact with the inner surface of the hollow part and the other side that comes into contact with the outer surface of the piston body part, and may provide an elastic force to the piston body part.
(17) A slit hole part penetrating an outer circumferential surface of the housing part may be provided. The pedal simulator of the vehicle may further include a retainer part penetrating the slit hole part to be combined with the housing part and interfering with the piston pressurization part to prevent the separation of the piston part.
(18) The retainer part may be formed in a ring form with one open side.
(19) The piston part may further include a magnet part mounted on an outer circumferential surface of the piston body part and formed in a circumferential direction of the piston body part.
(20) The pedal simulator of the vehicle may further include a sensor part provided in the housing part and configured to detect a location of the magnet part.
(21) The damper part may be formed in a hollow form.
(22) The damper part may further include a protrusion part formed in the outer surface of the damper part in a way to protrude therefrom.
(23) The protrusion parts may be disposed in a plural number in the circumferential direction of the damper part by being spaced apart from each other.
(24) A pedal simulator of the vehicle according to the present disclosure includes a housing part detachably combined with a pedal part, a piston part slidably disposed in the housing part, and a damper part provided in the piston part and compressed onto the housing part by a contact thereto.
(25) The pedal simulator may further include a bracket part provided in the housing part and combined with the pedal part.
(26) The bracket parts may be disposed on the outer surface of the housing part in a plural number

by being spaced apart from each other.
(27) The present disclosure has an effect in that a pedal return spring can be deleted by the elastic part that elastically supports the piston part.
(28) Furthermore, the present disclosure has effects in that repair and replacement costs for the pedal simulator can be reduced and productivity can be improved because the pedal simulator of the present disclosure can be used in various types of pedal parts in common through the modulation of the pedal simulator which may be applied regardless of the type and shape of the pedal part.
(29) Furthermore, the present disclosure has an effect in that a pedal effort of the pedal simulator may be adjusted by lengths, external diameters, hardness, and the like of the housing part, the elastic part, and the damper part.
(30) Furthermore, the present disclosure has an effect of measuring a pedal stroke by a magnet part integrally provided in the piston part.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view illustrating the state in which a pedal simulator of a vehicle according to an embodiment of the present disclosure is mounted on a pendant type pedal part.
(2) FIG. **2** is an outside perspective view illustrating the pedal simulator of the vehicle according to an embodiment of the present disclosure when viewed from one direction.
(3) FIG. **3** is an outside perspective view illustrating the pedal simulator of the vehicle of FIG. **2** when viewed from another direction.
(4) FIG. **4** is an exploded perspective view illustrating the pedal simulator of the vehicle according to an embodiment of the present disclosure when viewed from one direction.
(5) FIG. **5** is an exploded perspective view illustrating the pedal simulator of the vehicle of FIG. **4** when viewed from another direction.
(6) FIG. **6** is a front view illustrating a damper part in the pedal simulator of the vehicle according to an embodiment of the present disclosure.
(7) FIG. **7** is a cross-sectional view illustrating initial braking of the pedal simulator of the vehicle according to an embodiment of the present disclosure.
(8) FIG. **8** is a cross-sectional view illustrating middle and latter braking of the pedal simulator of the vehicle according to an embodiment of the present disclosure.
DETAILED DESCRIPTION
(9) Hereinafter, a pedal simulator of a vehicle according to an embodiment of the present disclosure will be described Referring to the accompanying drawings. In this process, thicknesses of lines illustrated in the drawings or sizes of constituent elements may be exaggerated for clarity and convenience of description. Furthermore, the terms used below are defined in consideration of the functions thereof in the present disclosure and may vary depending on the intention of a user or an operator or a usual practice. Accordingly, the definition of the terms should be made based on the entire contents of the present specification.
(10) FIG. **1** is a perspective view illustrating a state in which a pedal simulator of a vehicle according to an embodiment of the present disclosure is mounted to a pendant type pedal part.
(11) Referring to FIG. **1**, a pedal simulator **1** of a vehicle according to an embodiment of the present disclosure may be detachably mounted regardless of the type and shape of a pedal part **10**, such as a pendant type pedal part or an organ type pedal part. Accordingly, the pedal simulator **1** of the present disclosure may be modulated, may be conveniently assembled and mounted on the pedal part **10**, such as a pendant type pedal part or an organ type pedal part, and may be used in common.

(12) In the pedal simulator **1** of the present disclosure, a bracket part **150** provided in a housing part **100** may be conveniently assembled by being combined with a pendant type pedal part or an organ type pedal part through the medium of a coupling member **20**, such as a bolt or a nut.
(13) FIG. **2** is an outside perspective view illustrating the pedal simulator of the vehicle according to an embodiment of the present disclosure when viewed from one direction. FIG. **3** is an outside perspective view illustrating the pedal simulator of the vehicle of FIG. **2** when viewed from another direction. FIG. **4** is an exploded perspective view illustrating the pedal simulator of the vehicle according to an embodiment of the present disclosure when viewed from one direction. FIG. **5** is an exploded perspective view illustrating the pedal simulator of the vehicle of FIG. **4** when viewed from another direction. FIG. **6** is a front view illustrating a damper part in the pedal simulator of the vehicle according to an embodiment of the present disclosure.
(14) Referring to FIGS. **2** to **8**, the pedal simulator **1** of the vehicle according to an embodiment of the present disclosure includes the housing part **100**, a piston part **200**, an elastic part **300**, a damper part **400**, and a cap part **500**, which are described in detail as follows.
(15) The housing part **100** may include a hollow part **110**, a screw part **120**, and a guide part **130**.
(16) The hollow part **110** is provided within the housing part **100**, and is formed in the form of a hollow having a set length. The piston part **200** described later may be movably received in the hollow part **110**.
(17) The hollow part **110** may provide guidance to the straight movement of a piston body part **210**. An opening that communicates with the hollow part **110** is provided on an outer surface (the left/right sides of FIG. **7**) of the housing part **100**. The hollow part **110** may be formed as a hole having a cylindrical shape.
(18) The hollow part **110** may include a first hollow part **111**, and a second hollow part **112** having the smaller inner diameter than the inner diameter of the first hollow part **111**. Accordingly, a height difference part, which is stepped in a curved manner toward the inside of the housing part **100**, may be formed between the first hollow part **111** and the second hollow part **112**. The elastic part **300** described later may be mounted on the height difference part described later.
(19) A first thread part **100***a*, to which the screw part **120** described later is screw-coupled, may be provided in the hollow part **110**. A screw thread is formed in the first thread part **100***a*. The first thread part **100***a* may be located on an inner circumferential surface of one side (the right side of FIG. **7**) of the second hollow part **112**.
(20) The screw part **120** may be screw-combined with the hollow part **110**. The screw part **120** may include a middle room part **121** inserted into the inside of the second hollow part **112**, a head part **122** integrally provided in the middle room part **121** and seated on an end of one side of the second hollow part **112**, and a second thread part **122***a* integrally provided between the middle room part **121** and the head part **122** and having an outer circumferential surface formed with the screw thread to be screw-combined with the first thread part **100***a*.
(21) The screw part **120**, which is screw-combined with the housing part **100**, may be moved forward or backward in an axial direction of the housing part **100**. The forward or backward movement of the screw part **120** in the axial direction may allow the adjustment of an effective stroke of the middle room part **121**. Furthermore, this movement may also adjust the contact point of the damper part **400**, described later, coming in contact with the middle room part **121** of the screw part **120**.
(22) The middle room part **121** may be formed in a middle room form having a predetermined length. Furthermore, the diameter of the second hollow part **112** may be formed to be greater than the diameter of the middle room part **121**. The diameter of the second thread part **122***a* may be formed to be greater than the diameter of the middle room part **121**. The diameter of the head part **122** may be formed to be greater than the diameter of the second thread part **122***a*.
(23) The guide part **130** is provided within the housing part **100**, and is provided on the outside of the middle room part **121**. Incidentally, the guide part **130** is provided between the housing part **100**

and the middle room part **121**, and is formed in the circumferential direction of the middle room part **121**. The guide part **130** may communicate with the first hollow part **111**, and may provide guidance to the straight movement of a piston rod part **230** described later.
(24) The housing part **100** may be detachably combined with the pedal part **10**, such as a pendant type pedal part or an organ type pedal part. The housing part **100** includes a bracket part **150** that is combined with the pedal part **10**. The bracket part **150** may be formed in the outer surface of the housing part **100** in a way to protrude therefrom, and may be provided in a plural number by being spaced apart from each other.
(25) The bracket part **150** may include a through hole part **151**. The through hole part **151** is formed to penetrate the bracket part **150**. The bracket part **150** is combined with a coupling hole that is formed in the pedal part **10** through the medium of the coupling member **20**, such as a bolt or a nut, so that the housing part **100** may maintain the state in which the housing part **100** has been stably combined with the pedal part **10** and the rotation of the housing part **100** may be prevented.
(26) A sensor part **140** may be further provided in the housing part **100**. The sensor part **140** may be mounted on an outer surface of the housing part. The sensor part **140** may be electrically connected to a controller (not illustrated) of a vehicle, and may detect a location of a magnet part **240** described later.
(27) The piston part **200** is slidably disposed in the housing part **100**. The piston part **200** may include a piston body part **210**, a piston pressurization part **220**, and a piston rod part **230**.
(28) The piston body part **210** is located within the housing part **100**. The piston body part **210** is disposed within the housing part **100**. The piston body part **210** may be received in the hollow part **110**. Incidentally, the piston body part **210** may be movably received in the first hollow part **111**. Incidentally, the piston body part **210** may be formed in the shape of a flat plate.
(29) A spherical ball part **211** may be provided in the piston body part **210**. The ball part **211** may be formed to protrude from an outer surface of the piston body part **210** toward the piston pressurization part **220** described later. A socket part **221** of the piston pressurization part **220** may be combined with the ball part **211**.
(30) The piston pressurization part **220** is mounted on the piston body part **210**. Incidentally, the piston pressurization part **220** is mounted on the ball part **211**. The piston pressurization part **220** is moved toward one side (the right side of FIG. **2**) when an external force is applied to the piston pressurization part **220**.
(31) The piston pressurization part **220** may be rotatably combined with the piston body part **210** in a joint manner. Incidentally, the socket part **221** provided in the piston pressurization part **220** may be rotatably combined with the ball part **211** provided in the piston body part **210**.
(32) Furthermore, the piston body part **210** may be combined with the piston pressurization part **220** by caulking. For example, the piston pressurization part **220** that is moved by an external force may maintain the state in which the piston pressurization part **220** is rotatably combined in the ball part **211** by pressurizing and caulking the opening of the socket part **221** toward the piston body part **210**. Furthermore, the piston body part **210** is caulked into the piston pressurization part **220**, so that an assembly time and cost may be reduced.
(33) The piston rod part **230** is provided on one side (the right side of FIG. **4**) of the piston body part **210**. Incidentally, the piston rod part **230** is formed in the outer surface of the piston body part **210**.
(34) The piston rod part **230** may be movably received in the hollow part **110**, and may be inserted into the guide part **130**. The diameter of the piston body part **210** may be formed to be greater than the diameter of the piston rod part **230**.
(35) An opening **230***a* may be formed at the end (the left side of FIG. **5**) of one side of the piston rod part **230**, which is directed toward the middle room part **121**. The piston rod part **230** may be formed in the form of a hollow. The piston rod part **230** may be formed in the form of a cylindrical shape having a set length. The damper part **400** described later and the cap part **500** may be

mounted on the piston rod part **230**.

(36) The magnet part **240** may be provided in the piston part **200**. Furthermore, the magnet part **240** may be provided on the outer circumferential surface of the piston body part **210**. The magnet part **240** may be inserted and ejected and integrally formed in the piston body part **210**.

(37) The magnet part **240** may be formed in the circumferential direction of the piston body part **210**. Accordingly, a location of the magnet part **240** may be detected regardless of a mounting location of the sensor part **140** mounted on the housing part **100**.

(38) The magnet part **240** may measure information on the location of the piston part **200**. The magnet part **240** is a magnet having a magnetic force, and may transmit a pedal effort that is pressurized by the piston part **200** or information on the location of the piston part **200** to the controller of the vehicle through the sensor part **140** based on a change in the magnetic field, which occurs as the magnet part **240** is moved along with the piston part **200**.

(39) The elastic part **300** elastically supports the piston part **200** within the housing part **100**. Incidentally, the elastic part **300** elastically supports the piston part **200** in the hollow part **110**.

(40) The elastic part **300** has one side (the right side of FIG. **7**) that comes into contact with the inner surface of the hollow part **110** and the other side (the left side of FIG. **7**) that comes into contact with the outer surface of the piston body part **210**, and provides an elastic force to the piston body part **210** that is moved by an external force applied to the piston pressurization part **220**.

(41) The elastic part **300** is interposed between the piston body part **210** and the second hollow part **112**, and is compressed by the piston body part **210** that is moved by an external force applied to the piston pressurization part **220**. The compressed elastic part **300** provides an elastic force (or an elastic restoring force) to the piston body part **210** so that the piston body part **210** returns to its original location. The elastic part **300** may be a coil spring that surrounds the outside of the piston rod part **230**.

(42) The damper part **400** is provided in the piston part **200**. Incidentally, the damper part **400** is received within the piston rod part **230**. Incidentally The damper part **400** may be fabricated to include an elastically deformable material, and may be received within the piston rod part **230** through the opening **230***a* of the piston rod part **230**. The end of one side of the damper part **400** may be exposed from the piston rod part **230** through the opening **230***a* of the piston rod part **230**.

(43) The damper part **400** may be conveniently assembled because the damper part **400** is combined with the piston part **200**. The damper part **400** may include rubber, silicon, or plastic as the elastically deformable material.

(44) The damper part **400** is moved along with the piston part **200** and is compressed onto the housing part **100** by a contact thereto. Incidentally, the damper part **400** comes into contact with or does not come into contact with the middle room part **121** depending on a moving direction of the piston part **200**.

(45) When the piston rod part **230** moves toward the screw part **120** by an external force that is applied to the piston pressurization part **220**, the damper part **400** comes into contact with the outer surface of the middle room part **121**. When the piston rod part **230** is inserted into the guide part **130**, the damper part **400** is compressed between the inner surface of the piston rod part **230** and the middle room part **121**.

(46) The damper part **400** may be formed in the form of a hollow the inside of which is empty. Openings that communicate with the internal space of the damper part **400** may be provided at both ends of the damper part **400**, respectively.

(47) The damper part **400** may further include a protrusion part **410**. The protrusion parts **410** may be formed in the outer surface of the damper part **400** in a way to protrude therefrom. The protrusion parts **410** may be provided at both ends of the damper part **400**, respectively.

(48) The protrusion parts **410** may be provided on one side (the left side of FIG. **7**) of the damper part **400**, which is directed toward the inside of the piston rod part **230**, and one side (the right side

of FIG. **7**) of the damper part, which is directed toward the middle room part **121**, respectively. The protrusion parts **410** may be disposed in a plural number in the circumferential direction of the damper part **400** by being spaced apart from each other.
(49) The cap part **500** is combined with the piston part **200**. The cap part **500** is formed in a hollow form with both open ends, and is interposed between the piston part **200** and the damper part **400**. The cap part **500** may include a contact part **510**, a non-contact part **520**, and a flange part **530**.
(50) The contact part **510** is inserted into the piston rod part **230** through the opening **230***a* of the piston rod part **230**. The contact part **510** may be interposed between an inner circumferential surface of the piston rod part **230** and the damper part **400**, and may come into contact with a first part **400***a* of the damper part **400**. The contact part **510** may restrict the deformation of the first part **400***a* when the damper part **400** is compressed and deformed.
(51) The non-contact part **520** is integrally provided in the contact part **510**, and, similarly to the contact part **510**, is inserted into the inside of the piston rod part **230** through the opening **230***a* of the piston rod part **230**. The inner diameter of the non-contact part **520** may be formed to be greater than the inner diameter of the contact part **510**. Accordingly, the non-contact part **520** may be spaced apart from a second part **400***b* of the damper part **400**, and may not come into contact with the second part **400***b* of the damper part **400**.
(52) The non-contact part **520** may allow the deformation of the second part **400***b* when the damper part **400** is compressed and deformed. Incidentally, the non-contact part **520** may allow the deformation of the second part **400***b* so that the volume of the second part **400***b* may extend when the damper part **400** is compressed and deformed.
(53) The flange part **530** may be seated on an end of one side (the right side of FIG. **7**) of the piston rod part **230**. The flange part **530** may be provided on an end of the non-contact part **520**, and may have the same diameter as the piston rod part **230**.
(54) The pedal simulator **1** of the vehicle according to an embodiment of the present disclosure may further include a retainer part **600**.
(55) The housing part **100** may be provided with a slit hole part **101**. The slit hole part **101** may be formed to penetrate the outer circumferential surface of the housing part **100**. The slit hole part **101** may include a first slit hole part **101***a* and a second slit hole part **101***b* that is located on the opposite side of the first slit hole part **101***a*.
(56) The retainer part **600** may be formed in a ring form with one open side. The retainer part **600** may pass through the first slit hole part **101***a*, and a free end of the retainer part **600** may be inserted into the inside of the second slit hole part **101***b* so that the retainer part **600** is combined with the housing part **100**.
(57) The retainer part **600** interferes with the piston pressurization part **220** to prevent axial separation of the piston part **200** through an opening of the housing part **100**.
(58) An operating process of the pedal simulator of the vehicle having the aforementioned construction according to an embodiment of the present disclosure is described as follows.
(59) FIG. **7** is a cross-sectional view illustrating initial braking of the pedal simulator of the vehicle according to an embodiment of the present disclosure. FIG. **8** is a cross-sectional view illustrating middle and latter braking of the pedal simulator of the vehicle according to an embodiment of the present disclosure.
(60) Referring to FIG. **7**, when the piston pressurization part **220** is pressurized by an external force, the piston rod part **230** is moved toward the middle room part **121**. The elastic part **300** that is seated on the inner surface of the first hollow part **111** by the movement of the piston rod part **230** is compressed and deformed by the pressurization of the piston body part **210**. When the elastic part **300** is compressed, a user may feel an initial braking sense.
(61) At this time, the location of the magnet part **240** moved by the piston part **200** is detected by the sensor part **140**. The sensor part **140** transmits information on the location of the piston part **200** or a pedal effort based on a change in the magnetic field to the controller of the vehicle.

(62) Referring to FIG. **8**, when the external force is continuously applied to the piston pressurization part **220** and the piston rod part **230** is inserted into the guide part **130**, the damper part **400** that is moved along with the piston rod part **230** is compressed onto the middle room part **121** by a contact, and the user may feel a middle and late braking sense.
(63) At this time, the location of the magnet part **240** moved by the piston part **200** is detected by the sensor part **140**. The sensor part **140** transmits information on the location of the piston part **200** or a pedal effort based on a change in the magnetic field to the controller of the vehicle. When the external force applied to the piston pressurization part **220** is released, the compressed elastic part **300** provides an elastic force (or an elastic restoring force) to the piston body part **210** so that the piston body part **210** returns to its original location.
(64) The pedal simulator **1** of the vehicle according to an embodiment of the present disclosure may not include a pedal return spring by the elastic part **300** that elastically supports the piston part **200**.
(65) The pedal simulator **1** of the vehicle according to an embodiment of the present disclosure may reduce repair and replacement costs for a product and improve productivity because the pedal simulator **1** of the vehicle may be used in various types of pedal parts **10** in common through the modulation of the pedal simulator **1**, which may be applied regardless of the type and shape of the pedal parts **10**.
(66) The pedal simulator **1** of the vehicle according to an embodiment of the present disclosure may adjust a pedal effort by lengths, external diameters, hardness, and the like of the housing part **100**, the elastic part **300**, and the damper part **400**.
(67) The pedal simulator **1** of the vehicle according to an embodiment of the present disclosure may measure a pedal stroke by the magnet part **240** that is integrally provided in the piston part **200**.
(68) Although exemplary embodiments of the disclosure have been disclosed for illustrative purposes, those skilled in the art will appreciate that various modifications, additions and substitutions are possible, without departing from the scope and spirit of the disclosure as defined in the accompanying claims. Thus, the true technical scope of the disclosure should be defined by the following claims.

Claims

1. A pedal simulator apparatus of a vehicle, the apparatus comprising: a housing part; a piston part slidably disposed in the housing part; an elastic part elastically supporting the piston part within the housing part; a damper part provided within the piston part including a piston body part, a piston pressurization part, and a piston rod part, and compressed onto the housing part by a contact thereto; and a hollow cap part combined with the piston part and interposed between the piston part and the damper part.
2. The pedal simulator apparatus of claim 1, wherein: the piston body part is disposed within the housing part and provided with a ball part; the piston pressurization part is disposed on a first side of the piston body part and provided with a socket part rotatably combined with the ball part; and the piston rod part is mounted on a second side of the piston body part and onto which the damper part and the cap part are mounted.
3. The pedal simulator apparatus of claim 2, wherein the housing part includes: a hollow part in which the piston part is movably received; a screw part that is screw-combined with the hollow part and includes a middle room part with which the damper part selectively comes into contact depending on a moving direction of the piston part; and a guide part that is provided outside the middle room part, that communicates with the hollow part, and into which the piston rod part is inserted.
4. The pedal simulator apparatus of claim 3, wherein a diameter of the piston body part is greater

than a diameter of the piston rod part.
5. The pedal simulator apparatus of claim 4, wherein the elastic part includes a first side that comes into contact with an inner surface of the hollow part and a second side that comes into contact with an outer surface of the piston body part, and provides an elastic force to the piston body part.
6. The pedal simulator apparatus of claim 2, wherein the housing part includes a slit hole part penetrating an outer circumferential surface of the housing part, and wherein the pedal simulator apparatus further includes a retainer part passing through the slit hole part to be combined with the housing part, and interfering with the piston pressurization part to prevent separation of the piston part.
7. The pedal simulator apparatus of claim 6, wherein the retainer part is formed in a ring form having one open side.
8. The pedal simulator apparatus of claim 2, wherein the piston part further includes a magnet part mounted on an outer circumferential surface of the piston body part, and disposed in a circumferential direction of the piston body part.
9. The pedal simulator apparatus of claim 8, further including a sensor part provided in the housing part and configured to detect a location of the magnet part.
10. The pedal simulator apparatus of claim 2, wherein the piston rod part includes an opening at an end of one side thereof, and is formed in a hollow form.
11. The pedal simulator apparatus of claim 10, wherein the damper part is received within the piston rod part and includes an elastically deformable material.
12. The pedal simulator apparatus of claim 11, wherein the cap part includes: a contact part coming in contact with a first part of the damper part to restrict deformation of the first part; a non-contact part spaced apart from a second part of the damper part to allow deformation of the second part; and a flange part seated on an end of one side of the piston rod part.
13. The pedal simulator apparatus of claim 12, wherein the non-contact part is integrally provided in the contact part and an inner diameter of the non-contact part is greater than an inner diameter of the contact part.
14. The pedal simulator apparatus of claim 1, wherein the damper part is formed in a hollow form.
15. The pedal simulator apparatus of claim 14, wherein the damper part further includes a protrusion part formed in an outer surface of the damper part and protruding therefrom.
16. The pedal simulator apparatus of claim 15, wherein the protrusion part is in plural and the plurality of the protrusion parts are disposed in a circumferential direction of the damper part by being spaced apart from each other.
17. A pedal simulator apparatus of a vehicle, the apparatus comprising: a housing part detachably combined with a pedal part; a piston part slidably disposed in the housing part; and a damper part provided within the piston part including a piston body part, a piston pressurization part, and a piston rod part and compressed onto the housing part by a contact thereto.
18. The pedal simulator apparatus of claim 17, further including a bracket part provided in the housing part and combined with the pedal part.
19. The pedal simulator apparatus of claim 18, wherein the bracket part is provided in plural in the housing part and the plurality of bracket parts are disposed on an outer surface of the housing part by being spaced apart from each other.
20. The pedal simulator apparatus of claim 17, further including: a hollow cap part combined with the piston part and interposed between the piston part and the damper part, wherein the cap part includes: a contact part coming in contact with a first part of the damper part to restrict deformation of the first part; a non-contact part spaced apart from a second part of the damper part to allow deformation of the second part.